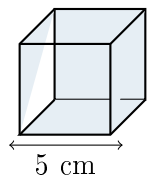


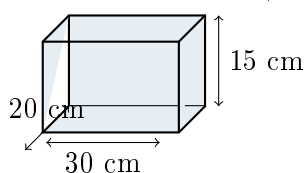
Atividade de Volume

Professor: Jefferson

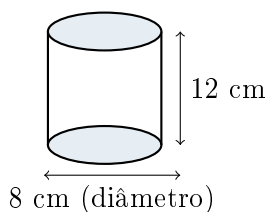
1. **Cubo:** Calcule o volume de um cubo com aresta de 5 cm.



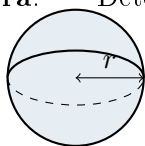
2. **Paralelepípedo:** Determine o volume desta caixa retangular em litros ($1L = 1000 \text{ cm}^3$).



3. **Cilindro:** Qual o volume deste cilindro? (Use $\pi = 3,14$)

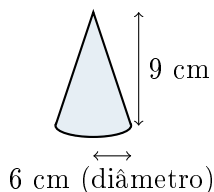


4. **Esfera:** Determine o raio r desta esfera.

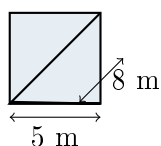


$$\text{Volume} = \frac{256\pi}{3} \text{ cm}^3$$

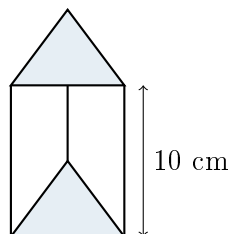
5. **Cone:** Calcule o volume deste cone.



6. **Pirâmide Quadrada:** Qual o volume desta pirâmide?

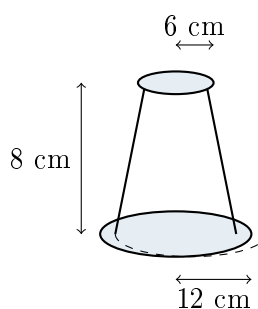


7. **Prisma Triangular:** Calcule o volume deste prisma.

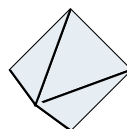


Base: triângulo 3-4-5 cm

8. **Tronco de Cone:** Determine o volume deste tronco de cone.

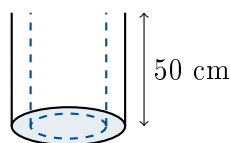


9. **Octaedro:** Calcule o volume deste octaedro regular.



Aresta: 4 cm

10. **Cilindro Oco:** Qual o volume de metal deste cano?



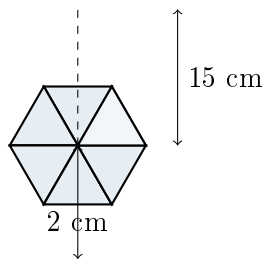
Externo: 6 cm
Interno: 4 cm

11. **Tetraedro:** Calcule o volume deste tetraedro regular.



Aresta: 6 cm

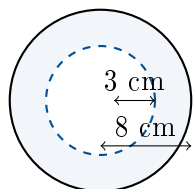
12. **Prisma Hexagonal Regular** Um prisma hexagonal regular possui aresta da base medindo 2 cm e altura de 15 cm, conforme ilustrado abaixo.



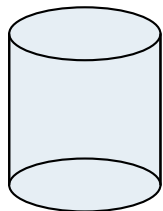
Determine:

- A área da base hexagonal
- O volume do prisma

13. **Anel Esférico:** Qual o volume da parte azul?



14. **Cilindro Inscrito:** Qual o volume do cilindro inscrito no cubo?

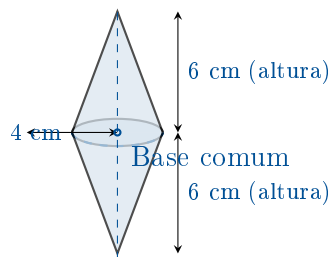


Cubo de aresta 4 cm

15. **Cones Gêmeos** Dois cones idênticos compartilham a mesma base circular. Sabendo que o raio da base é

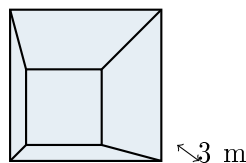
4 cm e a altura de cada cone é 6 cm, calcule o volume total da figura.

Eixo de simetria



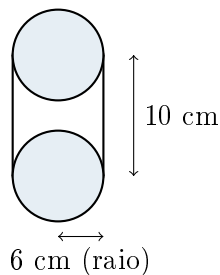
16. **Pirâmide Cortada:** Calcule o volume deste tronco de pirâmide.

Topo: 2 m $\times$ 2 m



Base: 4 m $\times$ 4 m

17. **Desafio: Sólido Complexo:** Calcule o volume deste sólido (cilindro + 2 semiesferas).



Fórmulas Úteis:

- | | |
|------------------------------------|---|
| • Cubo: $V = a^3$ | • Pirâmide: $V = \frac{1}{3}A_b h$ |
| • Paralelepípedo: $V = abc$ | • Tronco de cone: $V = \frac{\pi h}{3}(R^2 + Rr + r^2)$ |
| • Cilindro: $V = \pi r^2 h$ | • Octaedro: $V = \frac{\sqrt{2}}{3}a^3$ |
| • Cone: $V = \frac{1}{3}\pi r^2 h$ | • Prisma: $V = A_b \times h$ |
| • Esfera: $V = \frac{4}{3}\pi r^3$ | • Semiesfera: $V = \frac{2}{3}\pi r^3$ |